VISHWAKARMA AWARDS 2026–27 • ASHOKA UNIVERSITY MENTOR TRACK

IGNIS

Autonomous Dual-Domain Predictive Sentinel for Enclosed Tactical Firefighting Operations

DOMAIN	TARGET END-USER	PRIMARY INNOVATION	PROTOTYPE BUDGET
Physical AI & Resilience	**First Responders & Tactical Teams**	**Piezo-Acoustic & Heat Flux Fusion**	**₹24,500 / ₹30,000 Cap**

1. THE PROBLEM STATEMENT (PS)

In structural fires, internal search and rescue teams operate with zero visibility, subjected to two fatal, sudden hazards:

- **Thermal Flashover:** Unchecked thermal feedback drives ceiling heat over 500°C–600°C with heat flux > 20 kW/m², triggering instantaneous auto-ignition of all room contents.
- **Catastrophic Structural Shear:** Load-bearing composite timber and rebar fail microscopically long before visible sag.

The Void: Current hand-held Thermal Imaging Cameras (TICs) are passive, point-and-read visualizers requiring active firefighter line-of-sight. They offer **0s Predictive Warning** and zero structural vibration analysis.

2. THE ENGINEERED SOLUTION

IGNIS is an ultra-ruggedized, throwable/deployable autonomous sentinel entering enclosed compartments ahead of attack teams.

- **Dual-Domain Sensor Fusion:** Continuously correlates rapid non-contact thermal flux acceleration with contact acoustic stress wave emissions.
- **30–45s Pre-Flashover Alert:** Computes the second derivative of temperature (d^2T/dt^2) to trigger acoustic sirens and optical strobes before physical ignition occurs.
- **Sub-GHz NLOS Telemetry:** Uses 868 MHz peer-to-peer RF to cut through wet concrete walls, doubling as a tactical egress beacon.

3. NOVELTY & TECHNICAL DIFFERENTIATORS

Feature Dimension	Traditional Tech (TICs / Standard Gas Detectors)	IGNIS Autonomous Sentinel
Detection Principle	Static absolute threshold alert (alarms trigger after lethal heat is present).	**Predictive Differential Flux**: Triggers on thermal acceleration rate (dT/dt > 12°C/s).
Structural Integrity	None. Completely blind to internal beam shear stress and micro-fissures.	**Piezo-Acoustic Emission**: FFT analysis detects 20kHz–80kHz stress waves from wood/concrete fracture.
Tactical Form Factor	Handheld unit requiring dedicated operator attention in zero-visibility smoke.	Deployable autonomous node with 360° optical/ acoustic exit beacon and RF telemetry.
Signal Propagation	2.4 GHz Wi-Fi / Bluetooth (fails across wet concrete and structural walls).	Sub-GHz 868 MHz Non-Line-Of-Sight (NLOS) LoRa P2P link (penetrates up to 4 dense barriers).

4. HARDWARE STACK & GRANT ALLOCATION (₹30,000 BUDGET)

SYSTEM COMPONENT BREAKDOWN

THERMAL HARDENING & SENSING PHYSICS

Silica Aerogel Thermal Shielding: Internal components are encased inside an aerogel blanket

Subsystem	Component Specification	Cost (INR)
Compute / DSP	STM32H743 / ESP32-S3 Dual-Core MCU	₹2,800
Thermal Array	Melexis MLX90640 (32x24 Far-IR Sensor)	₹6,200
Acoustic Sensing	Wideband Contact Piezo Transducer + Preamp	₹2,100
RF Telemetry	Semtech SX1262 Sub-GHz 868MHz Transceiver	₹1,600
Enclosure / Shell	Sintered PA-CF Nylon + Silica Aerogel Core	₹5,500
Power & Warning	LiFePO4 High-Temp Cell + 100W Xenon Strobe	₹4,300
Total Estimated	**Complete Functional Field Prototype**	**₹22,500**

barrier (thermal conductivity $k < 0.015\ W/m{\cdot}K$), ensuring internal MCU temperatures remain below 65°C during 10-minute exposures at 250°C.

Mathematical Predictive Trigger:

$$Alert_{Flashover} = \left(\frac{dT}{dt} > \alpha \right) \wedge \left(\frac{d^2T}{dt^2} > 0 \right) \wedge \left(T_{ambient} > 180^\circ C \right)$$

A second acoustic vector analyzes continuous RMS acoustic emission energy in the micro-cracking band (30 kHz–60 kHz), flagging rapid brittle shear prior to structural collapse.

5. MASTER PRESENTATION DECK ARCHITECTURE (8 SLIDES)

Slide #	Title	Deliverables, Data Points & Visual Strategy
Slide 01	Title & Hook	High-contrast render of the ruggedized IGNIS sentinel. Subtitle: *Autonomous Pre-Flashover & Acoustic Structural Collapse Sentinel*. Team profile and Vishwakarma track identification.
Slide 02	The Fatal Blindspot	NIOSH / NFPA fatality statistics. Visual comparison showing why handheld TICs fail to alert firefighters during sudden flashover thermal runaways.
Slide 03	System Architecture	Complete schematic block diagram: IR thermopile array, piezo acoustic transducer, onboard edge MCU (FFT computation), and 868 MHz LoRa telemetry module.
Slide 04	The Deep Physics (Novelty)	Differential thermal flux curves (dT/dt) alongside high-frequency acoustic emission signatures. Explains the transition from reactive detection to predictive alerting.
Slide 05	Hardware & Thermal Packaging	Exploded CAD view illustrating the dual-wall carbon-fiber nylon shell, silica aerogel insulation layer, shock-damped suspension, and omnidirectional optical strobe.
Slide 06	Live Demonstration Protocol	Benchtop verification diagram: small-scale thermal chamber + mechanical stress rig demonstrating acoustic crack detection and heat-flux spike warning in real time.
Slide 07	Tactical Concept of Operations (CONOPS)	Step-by-step deployment timeline: Throw/place pod at entry threshold → Real-time NLOS telemetry to Incident Commander wrist unit → Automatic strobe egress beacon activation.
Slide 08	Commercialization & IP Roadmap	Unit economics at volume production (<₹6,500/node), utility patent claims on multi-modal flashover prediction, and engagement roadmap with regional fire departments.

6. WINNING LIVE DEMONSTRATION SCRIPT (FOR STAGE JURY)

Physical Stage Rig:

- A tabletop transparent heat enclosure equipped with a ceramic radiative heat element.
- A pre-stressed timber specimen mounted to a hand-cranked clamp with an integrated contact piezo probe.

Live Stage Sequence (2-Minute Impact):

- **00:30:** Heat element energized; IGNIS displays real-time telemetry graph of temperature and dT/dt.
- **01:15:** Heat acceleration surpasses safety gradient; IGNIS flashes its 360° strobe and sounds alert *well before* timber auto-ignites.
- **01:45:** Slight torque applied to timber; acoustic FFT spikes, flagging structural fracture before visual failure.